Hardware Object Drag and Adjustment Evidence

Generated 2026-06-02 13:38 KST

Gate interpretation correction. The visual-move screenshot is not proof that a hardware placement gate accepted the moved layout. It is a dirty visual preview: canonical CircuitSpec and verified wire routes remain unchanged, and Run/current controls stay disabled until reset or hardware placement resolution. Gate-pass evidence is the hardware-move resolver output plus the adjustment-mode E2E matrix.

Implementation Evidence

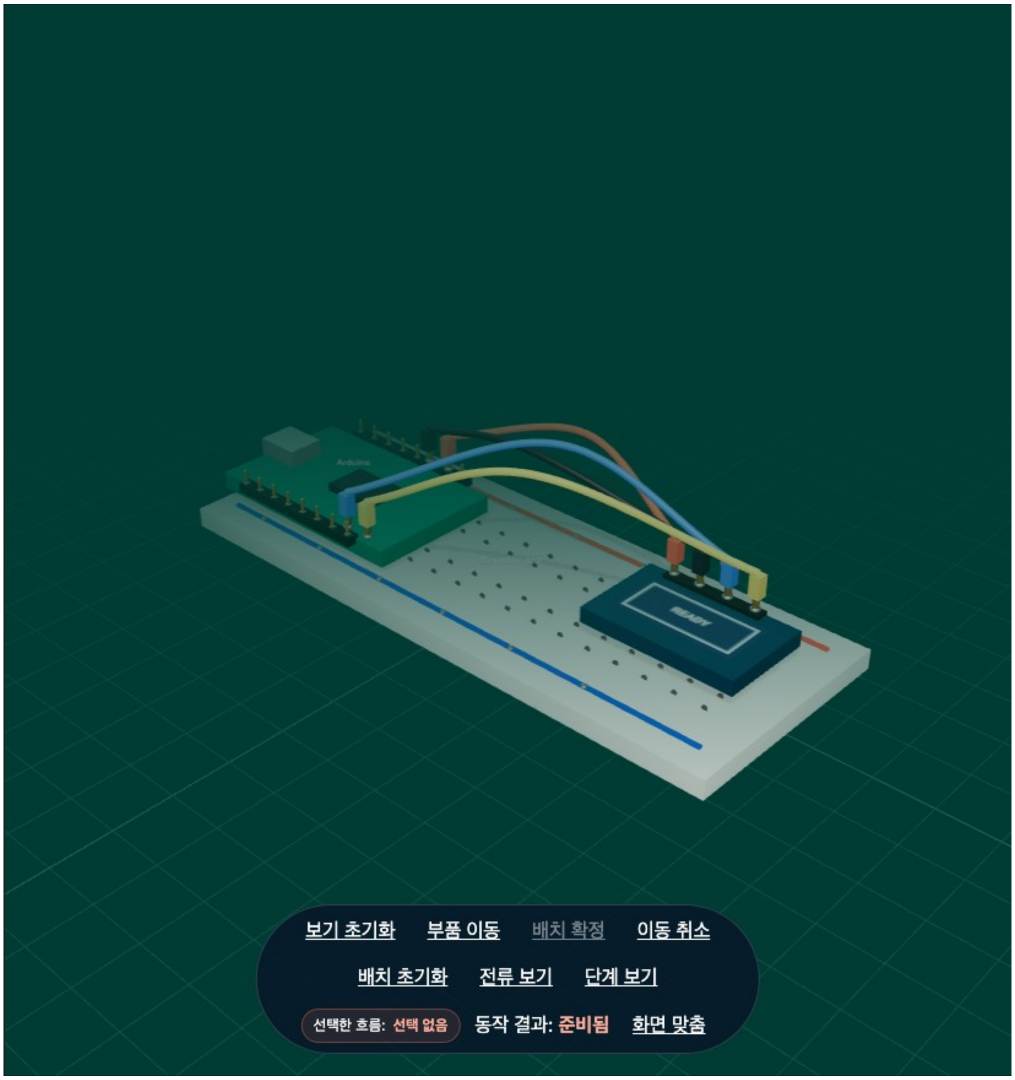
Scope	What Changed	Evidence	Status
Phase 1	Rule gate as automatic presentation adjustment	Four E2E adjustment-mode cases remain visible without unsupported controls.	Passed
Phase 2	Scene graph and debug snapshot	Part, wire, label, and preview-wire groups expose deterministic debug metadata.	Passed
Phase 3	Visual-only movement	Visual transforms move rendered objects and dashed preview wires without mutating CircuitSpec.	Passed
Phase 4	Hardware movement resolver	Dragged component is sent to placement API, snapped, rerouted, and returned as canonical artifacts.	Passed
QA fix	Library-only parts excluded from default PCB stage	Demo circuit no longer shows non-circuit library parts as if they were verified hardware.	Passed

Adjustment Mode Matrix

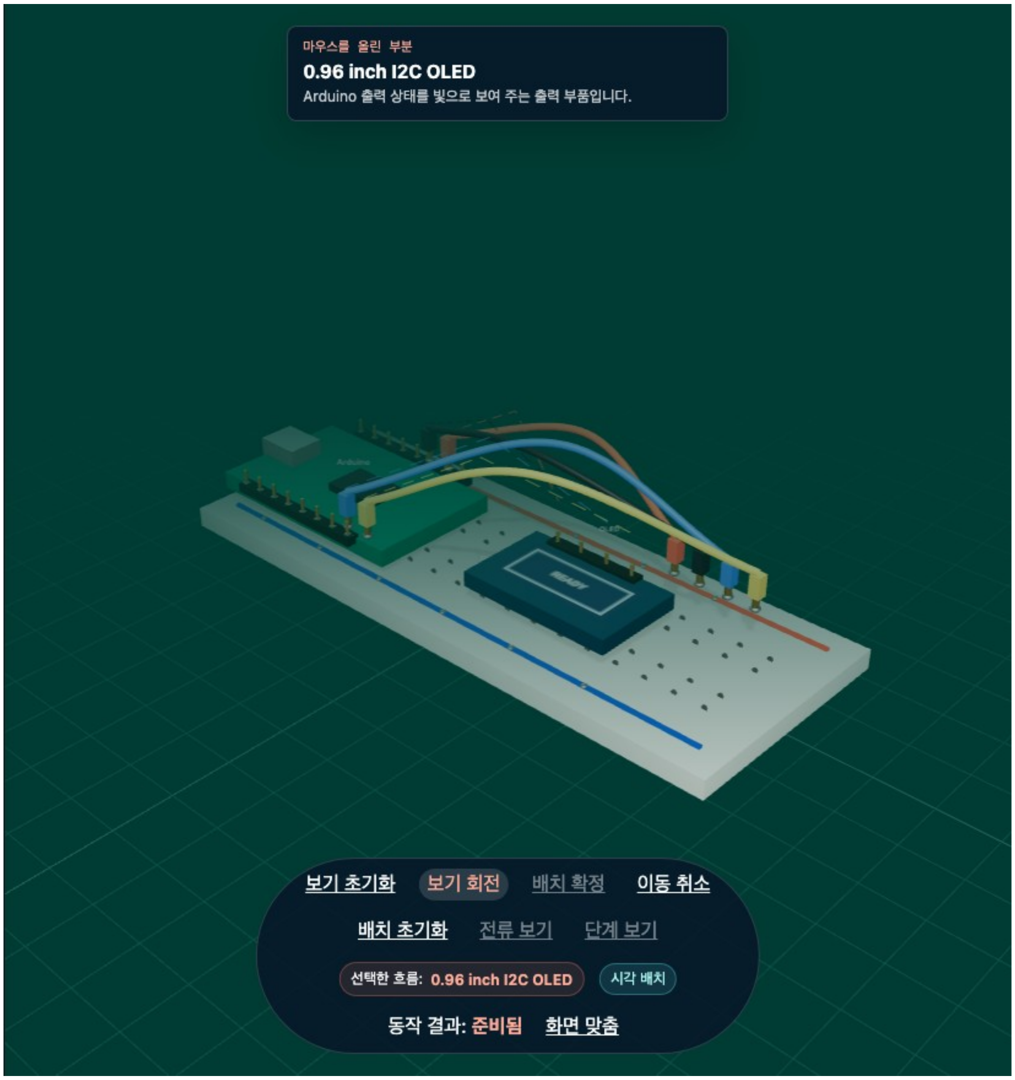
Mode	Visible Adjustment	Control Policy	Proof
Diagnostic	Safe 3D diagnostic scene remains visible while build/current claims stay scoped.	Run/current disabled unless verified paths exist.	Phase 1 E2E
Placeholder	Generic registered placeholder may render when exact footprint evidence is missing.	Exact geometry and pin claims disabled.	Phase 1 E2E
Safe Equivalent	Unsafe original is replaced with a safe displayed equivalent.	Original never build-ready; equivalent controls follow its own validation.	Phase 1 E2E
State Only	Pin-map, layout, and static state context can render without fake current flow.	Current animation off; static inspection on.	Phase 1 E2E

Browser Evidence Screenshots

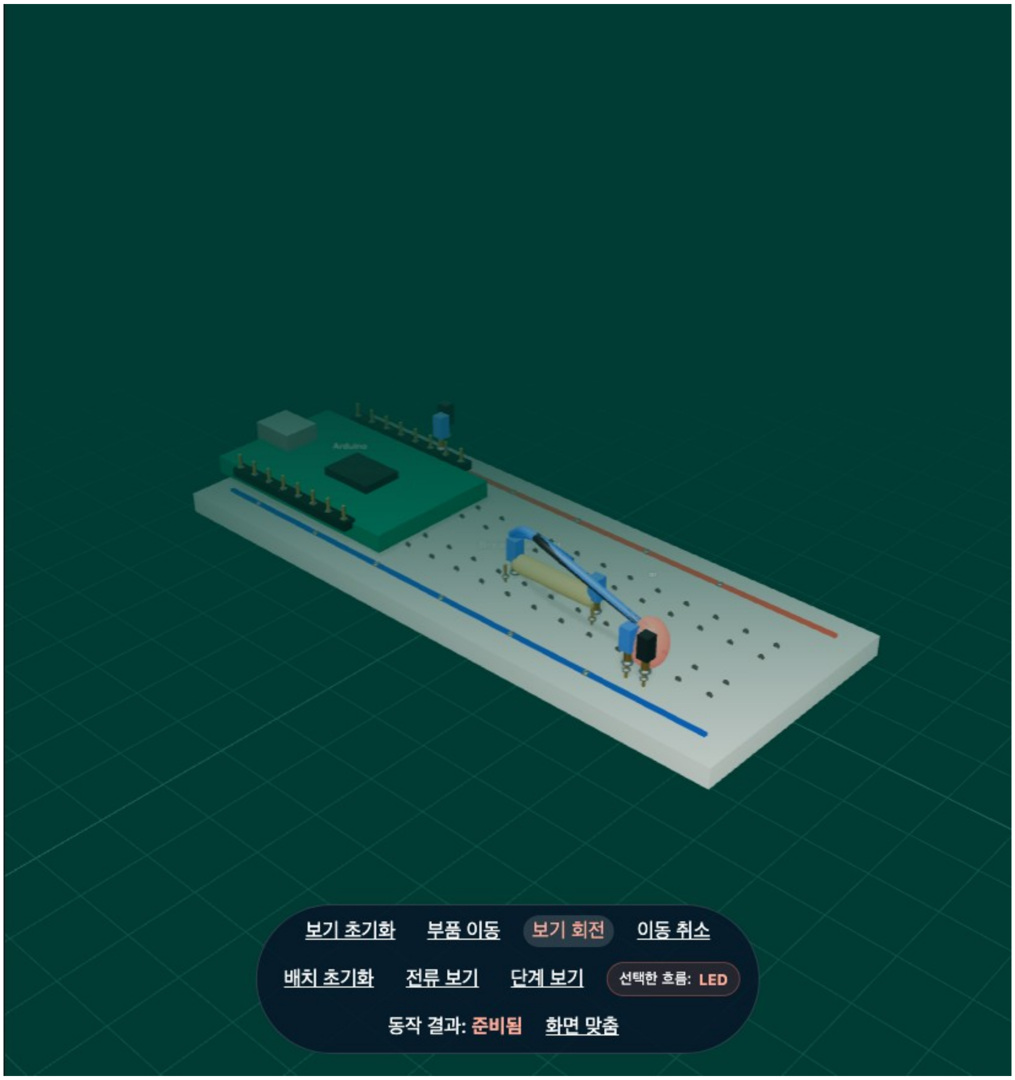
Baseline verified OLED demo scene. Library-only preview parts are no longer rendered in the PCB stage.



Visual-only movement preview. This is intentionally not a hardware gate pass; the scene is marked as visual arrangement and Run/current controls are disabled.



Hardware placement resolution. The dragged LED layout is returned through the placement API as a canonical adjusted render plan.



Verification Commands

Gate	Command or Case	Result
Syntax	node --check src/main.js && node --check src/stageScene.js	Passed
Typecheck	npm run typecheck	Passed
Unit	npm run test:unit	318 passed
Build	npm run build	Passed; existing Vite large chunk warning

		only
Targeted E2E	Phase 1 adjustment matrix, 3D render, visual move, hardware move	7 passed
Focused post-QA E2E	3D render, visual move, hardware move after libraryOnly fix	3 passed
Evidence capture	node scripts/generate_hardware_evidence.mjs	Passed

QA Notes

- The image that showed an off-board blue part was not a hardware gate-pass result; it was a visual preview and also exposed a libraryOnly rendering problem.
- Default PCB rendering now excludes libraryOnly parts so the visible stage matches the verified circuit under inspection.
- Visual movement is still allowed for inspection, but it stays dirty and disables Run/current controls until reset.
- Hardware movement is the canonical path: pointer-up sends a deterministic placement intent, then the server returns adjusted artifacts.
- The post-QA E2E test now asserts that hardware move changes wireRouteHash and remains buildReady after resolver output.